METR — MARCH 2025

Measuring AI Ability to Complete Long Software Tasks: Time Horizon Doubling Every 7 Months

Thomas Kwa, Ben West, Joel Becker et al.
Model Evaluation & Threat Research (METR) — arxiv.org/abs/2503.14499

~207d
DOUBLING TIME

R^2=0.97
TREND FIT

170
BENCHMARK TASKS

A New Metric: 50% Task-Completion Time Horizon

The task duration (in human time) at which an AI model achieves 50% success rate

Assemble Tasks

1

Curate 170 diverse software & research tasks spanning 3 seconds to 8 hours of human effort

3 suites • 800+ human baselines

→

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure 2 — Example tasks spanning seconds to hours of human effort

170 Tasks Spanning 3 Seconds to 8 Hours

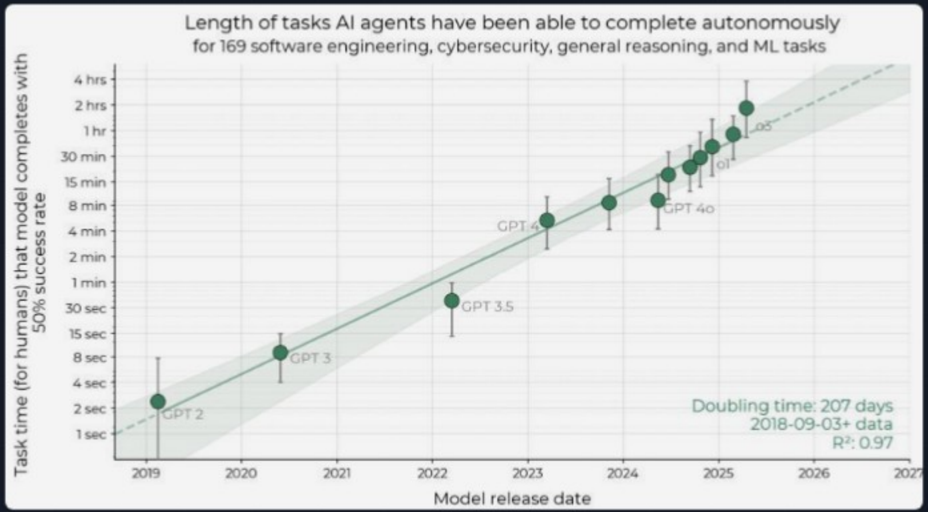
97	7	66
HCAST	**RE-Bench**	**SWAA**
Diverse software, 1 min – 30 hrs	ML research engineering, 8 hrs each	Atomic actions, 1 sec – 30 sec

TASK FAMILY	HUMAN TIME	DESCRIPTION
find_shell_script	3 sec	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 min	Research simple factual information from Wikipedia
oxdna_simple	9 min	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 min	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hrs	Implement custom CUDA kernels for 30× speedup of a backtesting tool

Over 800 human baselines totaling 2,529 hours collected across all tasks

From 2 Seconds to ~2 Hours in 6 Years

50% time horizon vs. model release date, 2019–2025



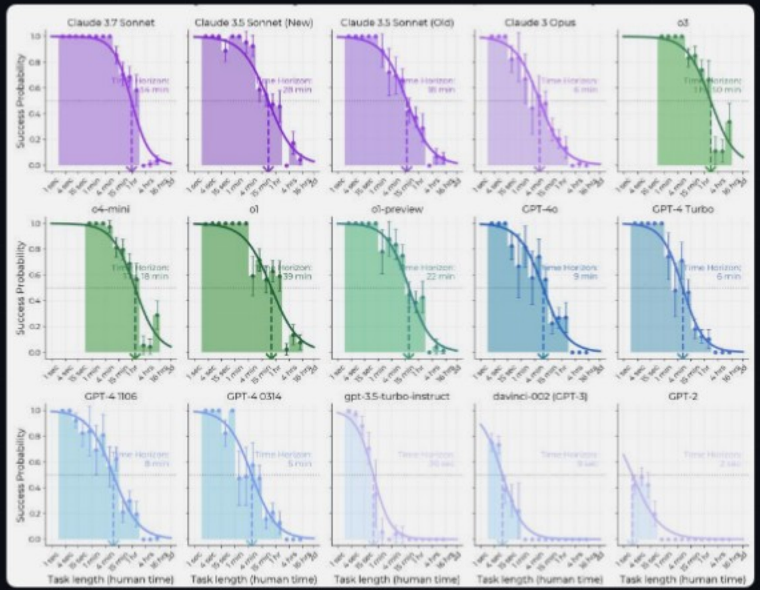
~207 days
DOUBLING TIME

R^2 = 0.97
EXPONENTIAL FIT

GPT-2 → o3
2 SEC → 1H50M

Success Drops Sharply as Task Length Increases

Per-model success probability curves — frontier models push the curve rightward



o3 (2025)
1h 50m
Highest 50% horizon

Claude 3.7 Sonnet
54 min
Strong frontier performer

o1
39 min
Reasoning-focused model

GPT-2 (2019)
~2 sec
Baseline starting point

Reasoning, Tool Use, and Error Recovery Drive Growth

Key capability drivers behind exponential time-horizon expansion

Improved Logical Reasoning

Models increasingly plan multi-step solutions, decompose problems, and maintain coherent strategies across longer task horizons

Better Tool Use

More effective use of code execution, file systems, and external APIs enables agents to interact with complex software environments

Error Recovery & Adaptation

Frontier models detect mistakes, adjust strategies, and recover from failures instead of getting stuck in loops

Reliability & Self-Awareness

Greater consistency in execution and ability to assess own confidence reduces cascading failures in long tasks

Month-Long Task Automation by 2028–2031

Forward extrapolation if the exponential trend continues

2019
GPT-2: ~2 seconds
Simplest atomic software actions

2023
GPT-4: ~5 minutes
Short coding tasks, debugging

2025
o3: ~1 hour 50 min
Complex multi-step engineering

2028–2031
Projected: ~1 month (167 work hrs)
Full month-long software projects

167 hrs

1-Month Work Horizon

Naive extrapolation predicts AI agents will autonomously complete tasks equivalent to a full month of skilled human work between mid-2028 and mid-2031

Possible Acceleration

The 2023–2025 growth rate is approximately **20% faster** than the overall 2019–2025 rate, suggesting the trend may be accelerating

Caveats

Extrapolation assumes continued exponential scaling. Real-world task distributions, regulatory changes, and compute constraints could alter the trajectory

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1. **Robust metric:** The 50% task-completion time horizon provides an intuitive, human-grounded measure of AI autonomy with $R^2 = 0.97$ over 6 years of model releases
2. **Consistent doubling:** ~7-month doubling time with possible acceleration since 2024 — the 2023–2025 rate is ~20% faster than the overall trend
3. **Current frontier:** o3 autonomously completes tasks that take skilled humans ~2 hours, up from 2 seconds (GPT-2) in just 6 years
4. **External validity:** Tasks may not perfectly represent real-world intellectual labor — performance on unstructured tasks is notably weaker, and generalization to real-world job distributions remains uncertain